LABORATORY REPORT 8: DYNAMIC ROUTING (OSPF) CONFIGURATION

STUDENT INFORMATION

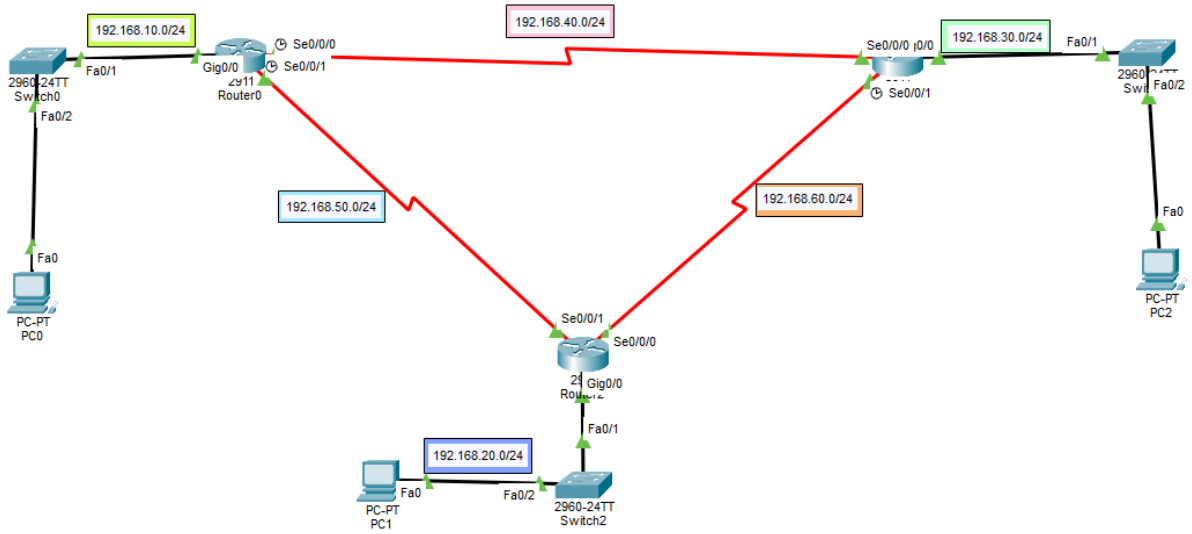
Item	Details
Student Name	Deepak Luitel
Enrollment Number	12250005
Date of Submission	16/05/2026

OBJECTIVE

To configure IP addresses on router interfaces, switches, and PCs to enable communication between devices on different networks using OSPF (Open Shortest Path First) as the dynamic routing protocol across multiple routers (2911 routers) in a multi-network topology.

IP ADDRESS DERIVATION

Design the network topology as shown below, but use unique numbers based on your own enrollment number.



Let:

X = Last two digits of enrollment number = ___05___

Network	Network Address	Router Interface IP
Network A (PC0 LAN)	192.168.5.0/24	192.168.5.1
Network B (PC1 LAN)	192.168.15.0/24	192.168.15.1
Network C (Router0 Gig0/0 to Switch0)	192.168.25.0/24	192.168.25.1
Network D (Router0 Se0/0/0 to Router1)	192.168.35.0/24	192.168.35.1 / .2
Network E (Router0 Se0/0/1 to Router2)	192.168.45.0/24	192.168.45.1 / .2
Network F (PC2 LAN on Router2)	192.168.55.0/24	192.168.55.1

Example: If Enrollment Number = 12250017 → X = 17

- Network A 192.168.17.0/24 PC0 (connected to Switch0 Fa0/2)
- Network B 192.168.27.0/24 PC1 (connected to Switch2 Fa0/2)
- Network C 192.168.37.0/24 Router0 Gig0/0 to Switch0 Fa0/1
- Network D 192.168.47.0/24 Router0 Se0/0/0 ↔ Router1 Se0/0/0
- Network E 192.168.57.0/24 Router0 Se0/0/1 ↔ Router2 Se0/0/0
- Network F 192.168.67.0/24 PC2 (connected to Switch3 Fa0/2)

IP ADDRESSING SCHEME

Device	Interface	IP Address	Subnet Mask	Gateway	Connected To
Router0	Gig0/0	192.168.25.1	255.255.255.0	N/A	Switch0 Fa0/1
	Se0/0/0 (DCE)	192.168.(X+30).1	255.255.255.0	N/A	Router1 Se0/0/0
	Se0/0/1 (DCE)	192.168.(X+40).1	255.255.255.0	N/A	Router2 Se0/0/0
Router1	Se0/0/0	192.168.(X+30).2	255.255.255.0	N/A	Router0 Se0/0/0
	Gig0/0	192.168.(X+10).1	255.255.255.0	N/A	Switch2 Fa0/1 (PC1)
Router2	Se0/0/0	192.168.(X+40).2	255.255.255.0	N/A	Router0 Se0/0/1
	Gig0/0	192.168.X.1	255.255.255.0	N/A	Switch0 Fa0/2 (PC0)

	Gig0/1	192.168.(X+50).1	255.255.255.0	N/A	Switch3 Fa0/1 (PC2)
PC0	NIC	192.168.X.2	255.255.255.0	192.168.X.1	Switch0 Fa0/2
PC1	NIC	192.168.(X+10).2	255.255.255.0	192.168.(X+10).1	Switch2 Fa0/2
PC2	NIC	192.168.(X+50).2	255.255.255.0	192.168.(X+50).1	Switch3 Fa0/2

ROUTER CONFIGURATION

Steps / Commands Used (GUI method in Packet Tracer):

Router0 Configuration

```
enable
configure terminal
hostname Router0

! Interface to Switch0 (192.168.(X+20).0/24)
interface gigabitethernet0/0
ip address 192.168.(X+20).1 255.255.255.0
no shutdown
exit

! Serial Link to Router1 (192.168.(X+30).0/24)
interface serial0/0/0
ip address 192.168.(X+30).1 255.255.255.0
clock rate 64000        ! DCE configuration
no shutdown
exit
```

```
! Serial Link to Router2 (192.168.(X+40).0/24)
interface serial0/0/1
ip address 192.168.(X+40).1 255.255.255.0
clock rate 64000        ! DCE configuration
no shutdown
exit

! OSPF Configuration
router ospf 1
router-id 1.1.1.1
network 192.168.5.0 0.0.0.255 area 0
network 192.168.35.0 0.0.0.255 area 0
network 192.168.55.0 0.0.0.255 area 0
end
write memory
```

Router1 Configuration

```
enable
configure terminal
hostname Router1

! Serial Link to Router0 (192.168.(X+30).0/24)
interface serial0/0/0
ip address 192.168.(X+30).2 255.255.255.0
no shutdown
exit

! Interface to Switch2 for PC1 (192.168.(X+10).0/24)
interface gigabitethernet0/0
ip address 192.168.(X+10).1 255.255.255.0
no shutdown
```

```
exit

! OSPF Configuration
router ospf 1
router-id 2.2.2.2
network 192.168.15.0 0.0.0.255 area 0
network 192.168.35.0 0.0.0.255 area 0
network 192.168.45.0 0.0.0.255 area 0
end
write memory
```

Router2 Configuration

```
enable
configure terminal
hostname Router2

! Serial Link to Router0 (192.168.(X+40).0/24)
interface serial0/0/0
ip address 192.168.(X+40).2 255.255.255.0
no shutdown
exit

! Interface to Switch0 for PC0 (192.168.X.0/24)
interface gigabitethernet0/0
ip address 192.168.X.1 255.255.255.0
no shutdown
exit

! Interface to Switch3 for PC2 (192.168.(X+50).0/24)
interface gigabitethernet0/1
ip address 192.168.(X+50).1 255.255.255.0
no shutdown
```

```
exit

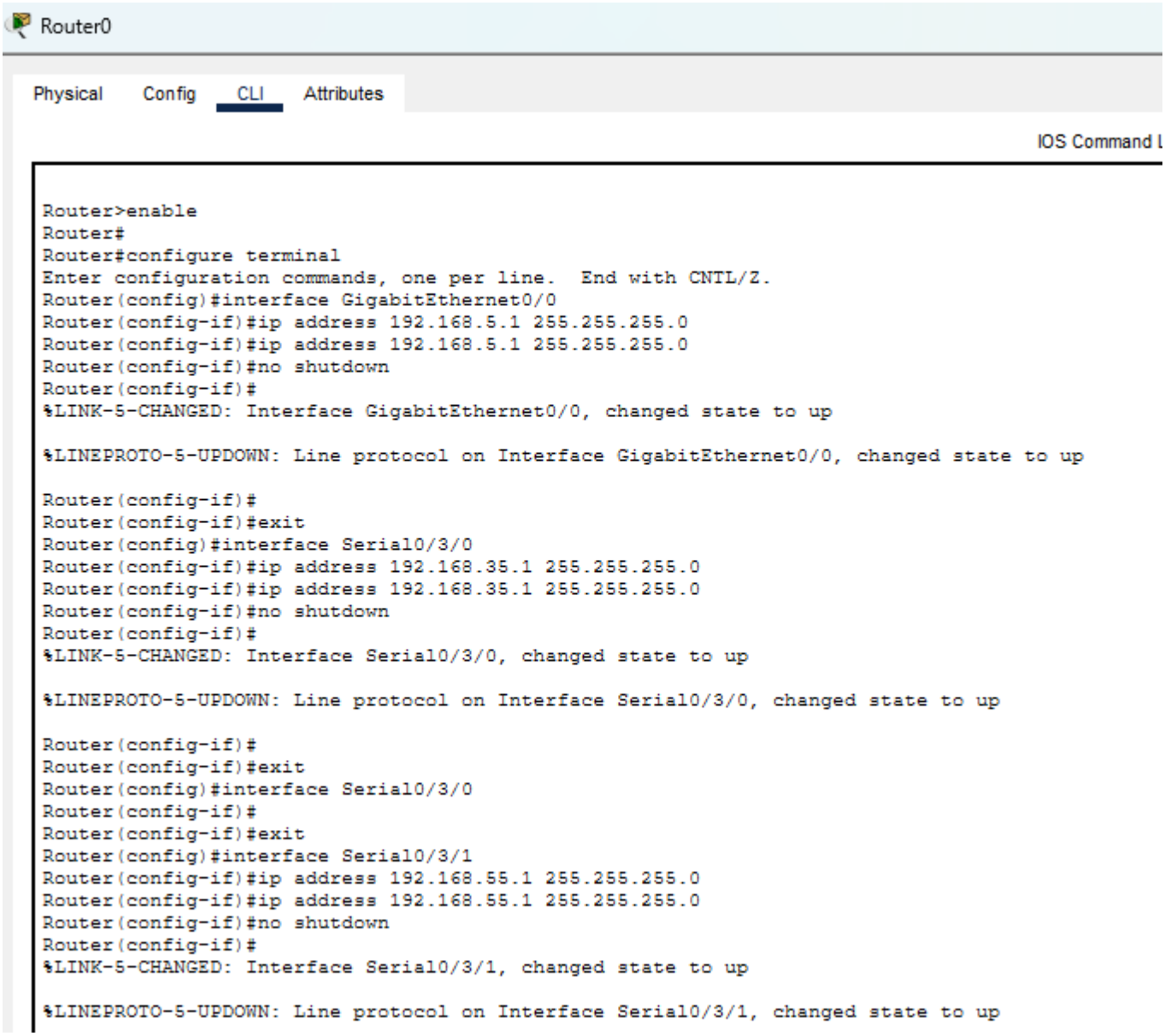
! OSPF Configuration
router ospf 1
router-id 3.3.3.3
network 192.168.25.0 0.0.0.255 area 0
network 192.168.55.0 0.0.0.255 area 0
network 192.168.45.0 0.0.0.255 area 0
end
write memory
```

Note: Replace X, X+10, X+20, X+30, X+40, X+50 with your actual calculated numbers.

Screenshot 1: Router Configuration

(Paste your router CLI configuration screenshot here)

Router 0:

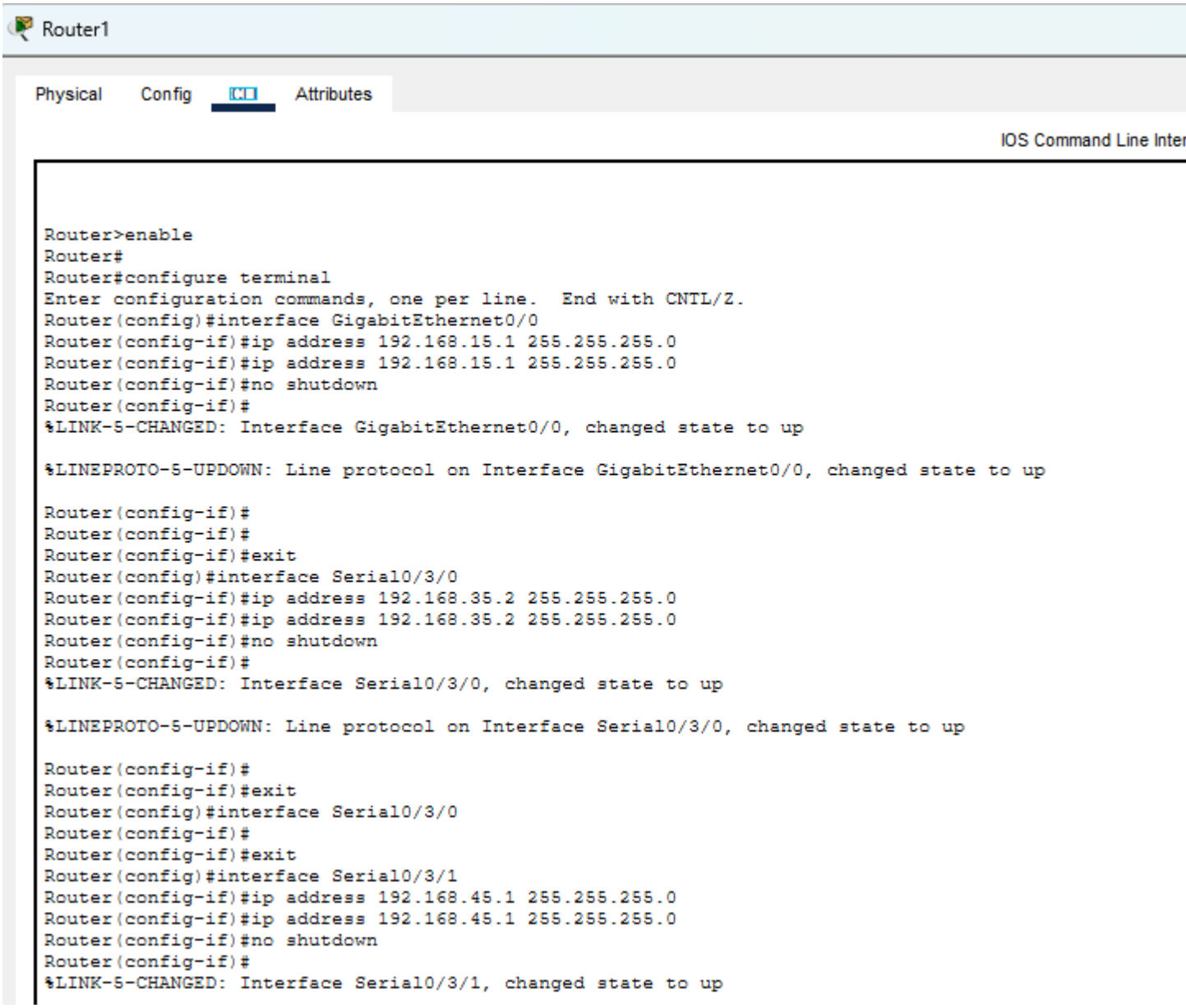

Router0
Physical
Config
CLI
Attributes
IOS Command L
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.5.1 255.255.255.0
Router(config-if)#ip address 192.168.5.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 192.168.35.1 255.255.255.0
Router(config-if)#ip address 192.168.35.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/1
Router(config-if)#ip address 192.168.55.1 255.255.255.0
Router(config-if)#ip address 192.168.55.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

```
Router(config-if)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.5.0 0.0.0.255 area 0
Router(config-router)#network 192.168.35.0 0.0.0.255 area 0
Router(config-router)#network 192.168.55.0 0.0.0.255 area 0
Router(config-router)#end
Router#write memory
Building configuration...
[OK]
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
00:14:29: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/3/0 from LOADING to FULL, Loading Done

00:15:09: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/3/1 from LOADING to FULL, Loading Done
```

Router 1:


Router1
Physical
Config
CLI
Attributes
IOS Command Line Inter
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.15.1 255.255.255.0
Router(config-if)#ip address 192.168.15.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 192.168.35.2 255.255.255.0
Router(config-if)#ip address 192.168.35.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/1
Router(config-if)#ip address 192.168.45.1 255.255.255.0
Router(config-if)#ip address 192.168.45.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network 192.168.15.0 0.0.0.255 area 0
Router(config-router)#network 192.168.35.0 0.0.0.255 area 0
Router(config-router)#network 192.168.45.0 0.0.0.255 area 0
Router(config-router)#end
Router#write memory
Building configuration...
[OK]
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:15:08: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial0/3/0 from LOADING to FULL, Loading Done

Router#
00:15:55: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/3/1 from LOADING to FULL, Loading Done
```

Router 2:

IOS Command Line Interf

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.25.1 255.255.255.0
Router(config-if)#ip address 192.168.25.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/1
Router(config-if)#ip address 192.168.45.2 255.255.255.0
Router(config-if)#ip address 192.168.45.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 192.168.55.2 255.255.255.0
Router(config-if)#ip address 192.168.55.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#router-id 3.3.3.3
Router(config-router)#network 192.168.25.0 0.0.0.255 area 0
Router(config-router)#network 192.168.55.0 0.0.0.255 area 0
Router(config-router)#network 192.168.45.0 0.0.0.255 area 0
Router(config-router)#end
Router#write memory
Building configuration...
[OK]
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:15:17: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial0/3/0 from LOADING to FULL, Loading Done

Router#
00:15:25: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/3/1 from LOADING to FULL, Loading Done
```

Top

VERIFICATION

Command: show ip route

Expected Output on Router0:

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2

Gateway of last resort is not set

C    192.168.(X+20).0/24 is directly connected, GigabitEthernet0/0
C    192.168.(X+30).0/24 is directly connected, Serial0/0/0
C    192.168.(X+40).0/24 is directly connected, Serial0/0/1
O    192.168.X.0/24 [110/65] via 192.168.(X+40).2, 00:00:04, Serial0/0/1
O    192.168.(X+10).0/24 [110/65] via 192.168.(X+30).2, 00:00:19,
Serial0/0/0
O    192.168.(X+50).0/24 [110/65] via 192.168.(X+40).2, 00:00:04,
```

```
Serial0/0/1
```

Screenshot of show ip route on **Router0**

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.5.0/24 is directly connected, GigabitEthernet0/0
L       192.168.5.1/32 is directly connected, GigabitEthernet0/0
O    192.168.15.0/24 [110/65] via 192.168.35.2, 00:06:09, Serial0/3/0
O    192.168.25.0/24 [110/65] via 192.168.55.2, 00:05:29, Serial0/3/1
     192.168.35.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.35.0/24 is directly connected, Serial0/3/0
L       192.168.35.1/32 is directly connected, Serial0/3/0
O    192.168.45.0/24 [110/128] via 192.168.35.2, 00:05:29, Serial0/3/0
                     [110/128] via 192.168.55.2, 00:05:29, Serial0/3/1
     192.168.55.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.55.0/24 is directly connected, Serial0/3/1
 --More--
```

Expected Output on Router1:

```
C    192.168.(X+30).0/24 is directly connected, Serial0/0/0
C    192.168.(X+10).0/24 is directly connected, GigabitEthernet0/0
O    192.168.(X+20).0/24 [110/65] via 192.168.(X+30).1, 00:00:03,
Serial0/0/0
O    192.168.(X+40).0/24 [110/65] via 192.168.(X+30).1, 00:00:03,
Serial0/0/0
O    192.168.X.0/24 [110/129] via 192.168.(X+30).1, 00:00:03, Serial0/0/0
O    192.168.(X+50).0/24 [110/129] via 192.168.(X+30).1, 00:00:03,
Serial0/0/0
```

Screenshot of show ip route on **Router1**

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    192.168.5.0/24 [110/65] via 192.168.35.1, 00:06:58, Serial0/3/0
     192.168.15.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.15.0/24 is directly connected, GigabitEthernet0/0
L       192.168.15.1/32 is directly connected, GigabitEthernet0/0
O    192.168.25.0/24 [110/65] via 192.168.45.2, 00:06:08, Serial0/3/1
     192.168.35.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.35.0/24 is directly connected, Serial0/3/0
L       192.168.35.2/32 is directly connected, Serial0/3/0
     192.168.45.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.45.0/24 is directly connected, Serial0/3/1
L       192.168.45.1/32 is directly connected, Serial0/3/1
O    192.168.55.0/24 [110/128] via 192.168.35.1, 00:06:08, Serial0/3/0
 --More--
```

Expected Output on Router2:

```
C    192.168.(X+40).0/24 is directly connected, Serial0/0/0
C    192.168.X.0/24 is directly connected, GigabitEthernet0/0
C    192.168.(X+50).0/24 is directly connected, GigabitEthernet0/1
O    192.168.(X+20).0/24 [110/65] via 192.168.(X+40).1, 00:00:12,
Serial0/0/0
O    192.168.(X+30).0/24 [110/65] via 192.168.(X+40).1, 00:00:12,
Serial0/0/0
O    192.168.(X+10).0/24 [110/129] via 192.168.(X+40).1, 00:00:12,
Serial0/0/0
```

Screenshot of show ip route on **Router2**

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    192.168.5.0/24 [110/65] via 192.168.55.1, 00:07:15, Serial0/3/0
O    192.168.15.0/24 [110/65] via 192.168.45.1, 00:07:05, Serial0/3/1
     192.168.25.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.25.0/24 is directly connected, GigabitEthernet0/0
L       192.168.25.1/32 is directly connected, GigabitEthernet0/0
O    192.168.35.0/24 [110/128] via 192.168.55.1, 00:07:05, Serial0/3/0
                     [110/128] via 192.168.45.1, 00:07:05, Serial0/3/1
     192.168.45.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.45.0/24 is directly connected, Serial0/3/1
L       192.168.45.2/32 is directly connected, Serial0/3/1
     192.168.55.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.55.0/24 is directly connected, Serial0/3/0
 --More-- |
```

Top

ADDITIONAL OSPF VERIFICATION COMMANDS

```
show ip ospf
show ip ospf neighbor
show ip ospf interface
show ip protocols
```

OSPF Verification

(Paste your OSPF verification screenshots here)

```
Router#show ip ospf
 Routing Process "ospf 1" with ID 1.1.1.1
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 3
        Area has no authentication
        SPF algorithm executed 4 times
        Area ranges are
        Number of LSA 3. Checksum Sum 0x014016
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

Router#show ip ospf interface

GigabitEthernet0/0 is up, line protocol is up
  Internet address is 192.168.5.1/24, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.5.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

```
% Invalid input detected at '^' marker.

Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -        00:00:32    192.168.35.2    Serial0/3/0
3.3.3.3           0   FULL/  -        00:00:32    192.168.55.2    Serial0/3/1
Router#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.5.0 0.0.0.255 area 0
    192.168.35.0 0.0.0.255 area 0
    192.168.55.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:10:34
    2.2.2.2              110      00:10:26
    3.3.3.3              110      00:10:26
  Distance: (default is 110)

Router#
```

CONNECTIVITY TESTING

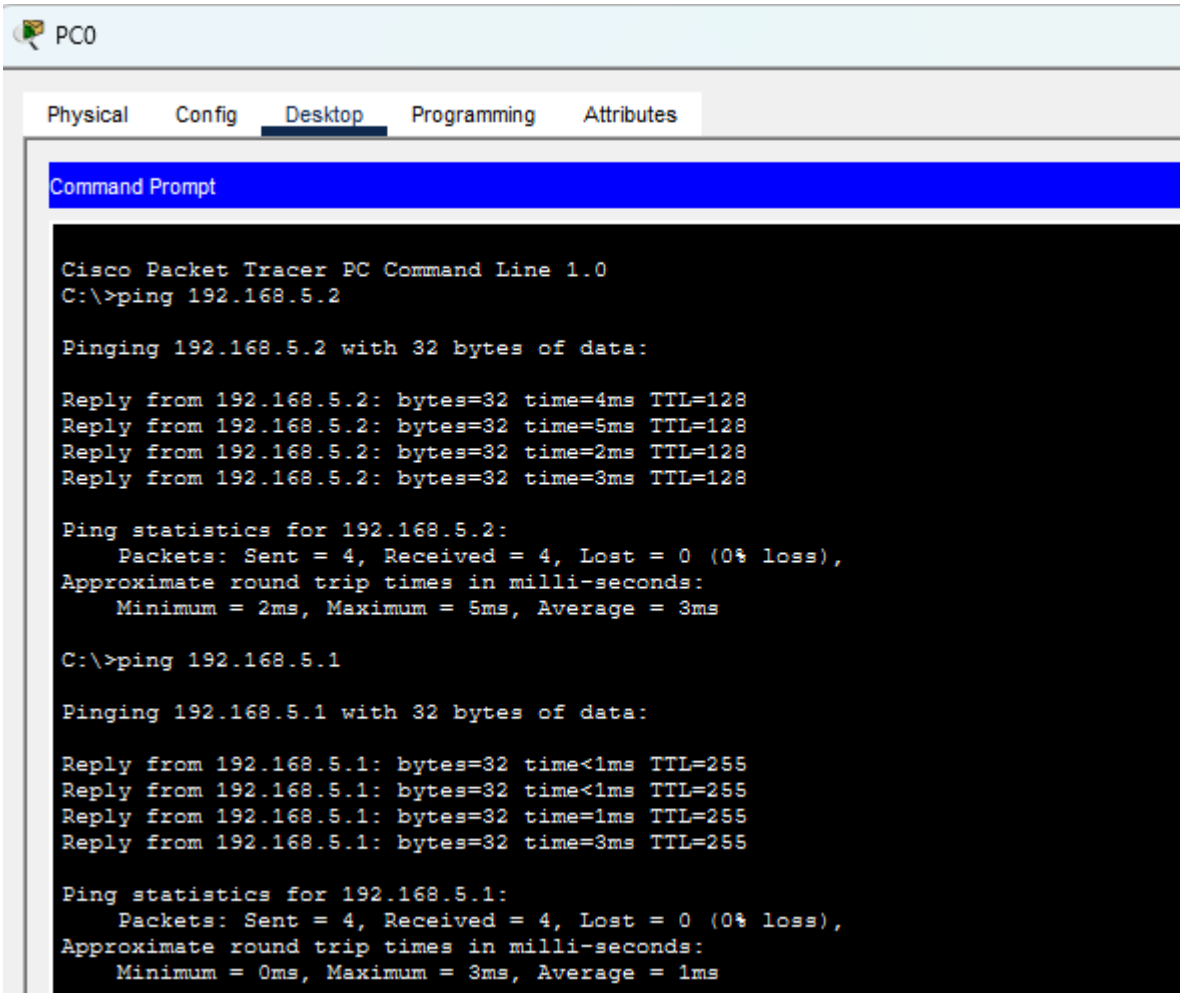
Test	Source	Destination	Command	Expected Result
Same network	PC0	PC0 (self)	ping 192.168.5.2	Successful
Same switch	PC0	Router2 Gig0/0	ping 192.168.5.1	Successful
Different routers (PC0 to PC1)	PC0	PC1	ping 192.168.15.2	Successful
Different routers (PC0 to PC2)	PC0	PC2	ping 192.168.25.2	Successful
Different routers (PC1 to PC2)	PC1	PC2	ping 192.168.25.2	Successful

Cross serial links	PC1	Router0 Se0/0/0	ping 192.168.35.1	Successful

Screenshot 4: Ping Test Results

(Paste your ping test screenshot here)

PC0 to PC 0 and PC0 to switch 0:



PC0 to PC1:

```
C:\>ping 192.168.15.2

Pinging 192.168.15.2 with 32 bytes of data:

Reply from 192.168.15.2: bytes=32 time=4ms TTL=126
Reply from 192.168.15.2: bytes=32 time=16ms TTL=126
Reply from 192.168.15.2: bytes=32 time=12ms TTL=126
Reply from 192.168.15.2: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.15.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 16ms, Average = 10ms

C:\>
```

PC 0 to PC2:

```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.25.2

Pinging 192.168.25.2 with 32 bytes of data:

Reply from 192.168.25.2: bytes=32 time=1ms TTL=126
Reply from 192.168.25.2: bytes=32 time=14ms TTL=126
Reply from 192.168.25.2: bytes=32 time=11ms TTL=126
Reply from 192.168.25.2: bytes=32 time=13ms TTL=126

Ping statistics for 192.168.25.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 14ms, Average = 9ms

C:\>
```

Top

PC 1 to PC2:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.25.2

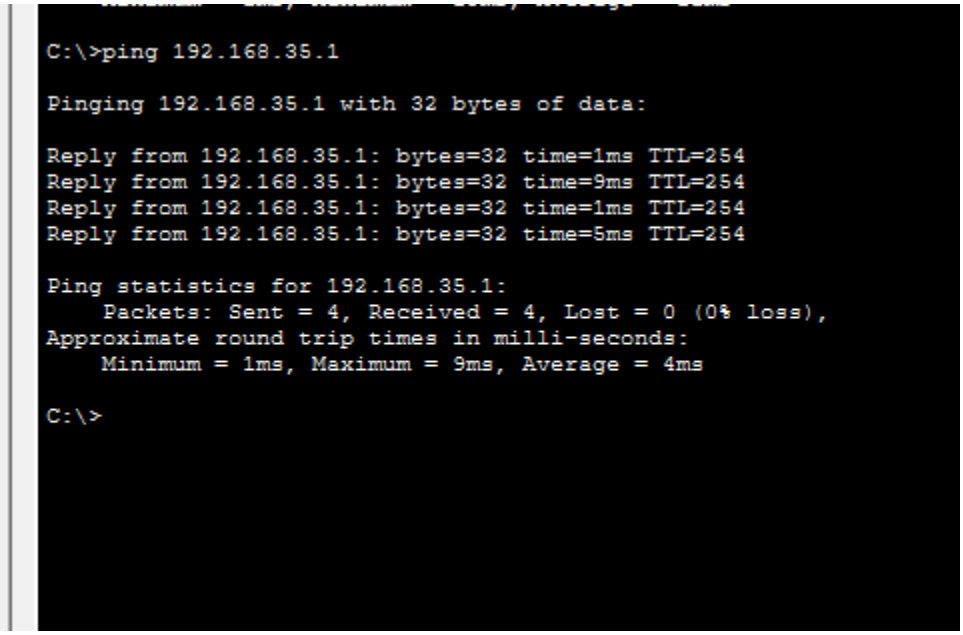
Pinging 192.168.25.2 with 32 bytes of data:

Reply from 192.168.25.2: bytes=32 time=1ms TTL=126
Reply from 192.168.25.2: bytes=32 time=19ms TTL=126
Reply from 192.168.25.2: bytes=32 time=10ms TTL=126
Reply from 192.168.25.2: bytes=32 time=14ms TTL=126

Ping statistics for 192.168.25.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 19ms, Average = 11ms

C:\>
```

PC 1 to Router 0 se0/0/0:



SHORT ANSWER QUESTIONS

1. **Why is OSPF considered a link-state routing protocol?**

OSPF is called a link-state routing protocol because routers share information about their links and network status with other routers to build a complete network map.

2. **What does "O" mean in show ip route output?**

“O” means the route was learned through OSPF routing protocol.

3. **What is the difference between OSPF and RIP?**

OSPF is faster, more scalable, and uses cost as a metric, while RIP is slower and uses hop count as a metric.

4. **What is the purpose of the router-id command in OSPF configuration?**

The router-id command gives a unique identification number to each router in the OSPF network.

5. **What is an OSPF area? Why is Area 0 used in this topology?**

An OSPF area is a logical group of routers used to organize the network. Area 0 is used because it is the backbone area required for OSPF communication.

CONCLUSION

Provide a concise summary (5–7 sentences):

In this lab, OSPF dynamic routing was successfully configured between multiple routers. IP addresses were assigned to routers and PCs for proper communication. OSPF helped routers automatically exchange routing information. Connectivity between different networks was verified using ping tests. The show ip route command confirmed that OSPF routes were learned correctly. This lab improved understanding of dynamic routing and OSPF configuration in Packet Tracer.

DECLARATION

I hereby declare that I have performed this laboratory work and all submitted evidence represents my own practical implementation.

Student Signature: _____D.L_____

Date: _____16/05/2026_____